

FABIO NEIRA ALZATE

GIS Infrastructure & Support Specialist | ArcGIS Enterprise · Python · Open Source · Cloud

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PROFESSIONAL SUMMARY

GIS Specialist and Environmental Engineer with 5+ years of experience in the administration, support, and maintenance of geospatial infrastructure in corporate environments. Proficient in ESRI platforms (ArcGIS Enterprise, ArcGIS Pro, ArcGIS Online), open source stack (QGIS, GeoServer), and cloud environments. Proven track record in web service publishing (MAP_IMAGE, FEATURE, TILE), Python/ArcPy automation, geodatabase administration, and Level 2 technical support for GIS users. Currently integrating AI agents and generative AI into geospatial workflows. Certified in leadership and team management. Committed to ensuring service availability, performance, and security while translating complex technical requirements into accessible solutions for business users.

CORE COMPETENCIES

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| • ArcGIS Enterprise Admin | • ArcGIS Online / Portal | • QGIS / GeoServer |
| • Python & ArcPy Automation | • GIS Web Service Publishing | • Enterprise Geodatabases |
| • Level 2 GIS User Support | • OGC Interoperability (WMS/WFS) | • Cloud Environments (AWS) |
| • Incident Diagnosis & Resolution | • AI Agents & Generative AI (GIS) | • Training & Knowledge Transfer |

PROFESSIONAL EXPERIENCE

GIS Infrastructure & Support Specialist

Jun 2021 – Present

HYG Consultores S.A.S. — Grupo SURA | Medellín, Colombia

- Administer and maintain a federated ArcGIS Enterprise 11.x environment, ensuring continuous availability of web GIS services (MAP_IMAGE, FEATURE, TILE) across 3+ business units of Grupo SURA.
- Automated the full service publishing lifecycle using Python Toolboxes (ArcPy), cutting deployment time by ~60% and eliminating manual errors in the process.
- Diagnose and resolve technical incidents in corporate GIS infrastructure, coordinating with IT teams to guarantee service continuity and geospatial data integrity.
- Designed and implemented a corporate Web GIS methodology based on OGC interoperability standards (WMS/WFS/REST), integrating ESRI services with open source platforms.
- Implemented OAuth2 Client Credentials flow for a public-facing geoportal, managing token lifecycle and service-sharing strategies across federated environments.
- Provide Level 2 technical support to GIS users, resolving configuration, service access, symbology, and performance issues across ArcGIS Pro, ArcGIS Online, and QGIS.
- Developed cartographic applications and web dashboards for non-specialist users, reducing the technical team's dependency for recurring operational queries.
- Led internal GIS training sessions designing learning plans adapted to each team's technical level under Scrum methodology.

GIS Instructor — University Level

Aug 2025 – Present

Politécnico Jaime Isaza Cadavid | Medellín, Colombia

- Design and deliver the GIS for Public Administration course, integrating open source tools (QGIS) and cloud platforms (ArcGIS Online, Google Earth Engine) for territorial management decision-making.
- Develop updated pedagogical materials, currently completing a Diploma in University Teaching to certify the design of organizational learning plans and training programs.

GIS Engineering Support Intern

Aug 2019 – Jun 2021

LINTEK S.A.S. | Colombia

- Processed and managed geographic databases for water and sewage infrastructure systems, supporting data loading and validation on the HydroCaz platform.
- Created, reprojected, and converted vector elements in GIS platforms, ensuring geometric and attribute consistency of delivered datasets.
- Handled topographic data and raster layers (DTMs, orthophotographs) for environmental and hydraulic infrastructure projects.
- Trained end users on the HydroCaz platform, ensuring effective tool adoption across technical teams.

EDUCATION

GIS Specialization (Graduate)

2024 – Jun 2025

Universidad de Manizales

- Advanced spatial analysis, Python automation, satellite imagery processing (ArcGIS Pro, QGIS, SNAP, Google Earth Engine). Thesis: SAR technology (Sentinel-1) for automated waterbody detection in coastal zones.

Environmental Engineering (B.Sc.)

Mar 2021

Universidad de Antioquia

- Complementary studies in environmental modelling, geomatics, risk management, water resource planning, and environmental auditing.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

AI Agents in Geospatial Projects

2025 – Present

In progress — Claude (Anthropic) · Google AI

- Hands-on training in designing and integrating AI agents into GIS solutions: workflow automation, AI-augmented spatial analysis, and LLM-powered GIS tools (Claude, Google Gemini).

Diploma in University Teaching

2025 – Present

In progress

- Certified to design organizational learning plans and structured training programs for technical and non-technical audiences.

Leadership & Team Management

Mar 2026

Universidad Internacional de La Rioja (UNIR)

- Team management, effective communication, decision-making, and leadership in technical and organizational environments. Credential ID: 77202603011143.

Generative AI Diploma

Jun 2025

Eidos Global + Microsoft

- Foundations and practical applications of generative artificial intelligence in professional and geospatial environments.

Advanced Geospatial Data Analytics in Python

Feb 2026

LinkedIn Learning

- Advanced spatial data analysis and visualization using Python for GIS workflows.

ArcGIS Systems Administration Diploma

Oct 2022

ESRI Colombia

- ArcGIS Enterprise configuration, branch versioning, geodatabase replication, multiuser workflows, and geospatial data management in production environments. Credential ID: 881222.

ArcGIS Enterprise: Administration Workflows

Apr 2024

ESRI Colombia

- Advanced enterprise GIS administration workflows, monitoring, and maintenance. ID: 16261.

Deploying & Maintaining a Multiuser Geodatabase

Aug 2022

ESRI Colombia

- Deployment, configuration, and maintenance of enterprise geodatabases in multiuser environments. IDs: 8736, 8750, 8752, 8902.

English B1 Certification

Dec 2025

Institución Universitaria ESUMER

- Certified English B1 proficiency with focus on academic and professional technical communication.

AWARDS & RECOGNITIONS

- USAID Leadership Scholarship — Open Data Mapping & Leadership. Training conducted in South Africa, 2019.
- YouthMappers Leadership Fellow — YouthMappers. Recognized for leadership in open data mapping and geospatial volunteering, Jan 2019.
- ACRES Foundation Scholarship — Cámara Colombiana de Infraestructura. Full academic merit scholarship, 2017–2021.
- Lead Member — GeoLab Research Group, Universidad de Antioquia. Geospatial university research and open innovation, 2017–2021.

LANGUAGES

Spanish:

Native

English:

Intermediate (B1 — ESUMER Certified)